

From Narrative to Alpha

Exogenous Perception Pressure as an Orthogonal Signal in an Autonomous Microstructure Alpha Book: Timescale Separation, Effective Breadth, and the Adverse-Selection Barrier

prism_signal_preprint.tex · 9pp · Jul 2026 · theory + protocol, **no performance figures by design**

The problem it names

A book built entirely from order-book microstructure is powerful but **structurally narrow**:

- one information source (order flow),
- one band of timescales (sub-second to minutes),
- jointly capped by an adverse-selection barrier that collapses a large unconditional IC once you condition on the states where a passive fill actually occurs (#2, #7).

Question: what does a *single exogenous signal from a different information domain* contribute to such a book? Answered at the level of methodology, not measured performance.

Integration architecture (source-agnostic)

A narrative series is aligned to market bars by a **point-in-time as-of join** and then scored by *the same* IC, MI, and five-gate replication machinery as any endogenous feature. The exogenous signal gets no privileges — that is the design’s credibility.

Three reasons it can help without being individually strong

I. Timescale separation. Narrative momentum lives in a spectral band essentially disjoint from the 0.005–0.1 Hz band that carries order-book-imbalance predictability (#7). The two are close to orthogonal *by construction*, not by luck.

II. Effective breadth. Under the fundamental law,

$$\text{IR} \approx \overline{\text{IC}}\sqrt{N_{\text{eff}}}, \quad N_{\text{eff}} = \frac{N}{1 + (N-1)\bar{\rho}}$$

appending a *low-correlation* signal raises N_{eff} more than appending another correlated microstructure signal *of equal strength*. This is the quantitative core: a weak orthogonal signal can beat a strong redundant one, and the formula says by how much.

III. Partial evasion of the adverse-selection barrier. A slow exogenous signal is harvested by *patient* execution over its own horizon rather than by reactive spread-crossing, so its coupling to the fillable-state indicator — the mechanism that destroys microstructure edge — is weaker. Its conditional IC need not collapse the same way.

Status of each claim

[!] All three are stated as **hypotheses with explicit, leakage-controlled, FDR-corrected tests**. The paper reports **no performance figures**. Reason III in particular is an argument about a mechanism, not a measurement — it is the one most likely to be attacked, and the one worth testing first.

What “more complete” means

The contribution is a theory and a protocol for making an autonomous microstructure research platform *more complete*: a decorrelated, multi-domain alpha book **that reports its own honest effective breadth**, admitting narrative as a first-class, gated signal source rather than an ad-hoc overlay.

Why this matters strategically

It is the argument for why the platform should not just keep mining order-flow features. Adding the $n+1$ -th imbalance variant moves

N_{eff} almost not at all once $\bar{\rho}$ is high; adding one genuinely exogenous input moves it materially. If the microstructure edge is execution-bounded (and #2/#7 say it is), **breadth is the remaining lever**.

The honest weak point

N_{eff} gains are computed with $\bar{\rho}$ estimated in-sample and in calm conditions. #6 makes the matching argument that **the binding correlation is the stress-regime one** — narrative and microstructure may correlate precisely during the events that matter. Any test must estimate $\bar{\rho}$ conditional on stress, or the breadth gain is an artifact.

The feature-source contract

Design principle: **the exogenous signal must be indistinguishable, to the scoring machinery, from an endogenous feature** — held to the same standard, no special pleading. A source exposes `load(symbol, start, end)` returning a frame indexed by UTC timestamp, one column per narrative component. The process runner merges onto market bars; thereafter a narrative column is scored by `ic_horizon` and `mi_ksg` exactly as an order-book column is. **Processes operate on columns and never learn a column’s origin** — narrative is first-class because it is admitted by the type system, not by a bespoke code path. Alignment is a **backward as-of join**: a bar sees only narrative stamped at or before its close.

Evaluation protocol — each with its kill criterion

- Orthogonality.** Volatility-regime-conditional ρ_x between narrative momentum and the incumbent imbalance signal, plus cross-spectral coherence. *Kill*: if ρ_x is not small in **stress regimes**, the breadth argument does not apply at all.
- Incremental content.** $I(r_{t \rightarrow t+h}; \dot{P}_t \mid \text{price}_t, \text{volume}_t)$ at narrative horizons, k NN estimator, block-permutation null, BH across concepts and horizons. *Kill*: not significantly positive after FDR $\Rightarrow$ narrative adds nothing beyond price and volume.
- Breadth realization.** Book N_{eff} and OOS IR with and without the signal, walk-forward. *Kill*: neither rises on inclusion.
- Execution realizability.** Schedule-based execution. *Kill*: conditional IC collapses the way the microstructure signals’ does.

Note the discipline: **the kill criteria are written before any number exists**, which is what stops (3) from becoming an exercise in finding a window where N_{eff} went up.

Siblings

#4 defines $P(K, t)$ and the Prism system that produces it (read first) · #6 supplies the N_{eff} mathematics and the estimation-hygiene caveat above · #2 and #7 supply the barrier this paper proposes to partially evade.

Send-with

Pair with #4 for Leippold (UZH/SFI). Alone it reads as theory without an instrument; #4 alone reads as an instrument without a purpose.